

0.6 Polynomials

Definition 0.6.1 — Polynomial

A **polynomial** over a field F in the indeterminate x is an expression of the form:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{k=0}^n a_k x^k$$

where $a_0, a_1, \dots, a_n \in F$ are called the **coefficients** of $p(x)$.

Definition 0.6.2 — Degree

Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a nonzero polynomial with $a_n \neq 0$. The **degree** of $p(x)$, denoted $\deg p(x)$, is n . The coefficient a_n is called the **leading coefficient**.

By convention, the zero polynomial $p(x) = 0$ has degree $-\infty$.

Definition 0.6.3 — Polynomial Ring

The set of all polynomials over a field F is denoted $F[x]$:

$$F[x] = \left\{ \sum_{k=0}^n a_k x^k : n \in \mathbb{N}, a_k \in F \right\}$$

This set forms a **ring** under polynomial addition and multiplication. (You can omit what ring means here.)

Example. (Elements of the Polynomial Ring)

The following are elements of $\mathbb{R}[x]$:

$$3x^2 - 2x + 1, \quad x^5 + \pi x^3 - \sqrt{2}, \quad 7, \quad 0$$

Note that constant polynomials (including 0) are also polynomials.

In linear algebra, we often work with polynomials of bounded degree.

Definition 0.6.4 — Polynomials of Degree at Most n

Let F be a field and $n \in \mathbb{N}$. The set of all polynomials over F of degree at most n is denoted:

$$F[x]_{\leq n} = \{p(x) \in F[x] : \deg p(x) \leq n\}$$